ROHAN KHATAVKAR

Arizona State University, Tempe, AZ 85287, USA
rkhatavk@asu.edu | Website | Google Scholar | LinkedIn

RESEARCH

- **Wearable Robotics:** Mechatronic and pneumatic system design; soft active devices; haptic feedback devices; pneumatic artificial muscles; variable-stiffness mechanisms; soft parallel elastic actuators.
- **Human-Robot Interaction and Biomechanics:** Gait kinematics and kinetics; postural stability; human-subject experiments; electromyography; force myography; inertial sensing.
- **Modeling and Intelligent Systems:** Physics-based pneumatic system modeling; data-driven real-time human intent detection.

ACADEMIC POSITIONS

Graduate Research Associate — **Jan. 2024–present**
Arizona State University — *Tempe, USA*
Advisors: Prof. Jiefeng Sun and Prof. Hyunglae Lee

Research Fellow — **Oct. 2021–Oct. 2023**
Indian Institute of Technology Delhi — *New Delhi, India*
Advisor: Prof. Deepak Joshi

Selected Research Contributions

- **Soft Back-Support Devices:** Developed passive variable-stiffness and semi-active soft back-support devices that provide task-adaptive assistance and improve biomechanical and postural stability outcomes.
- **Haptic Cueing Device for Parkinson's Disease:** Developed a wearable, human-in-the-loop vibrotactile cueing system for mitigating freezing of gait and improving toe clearance.
- **Electronic Differential Locking Mechanism:** Developed and prototyped a compact electromagnetic locking mechanism into automotive differential.

EDUCATION

Ph.D. Mechanical Engineering — **Expected Dec. 2028**
Arizona State University — *Tempe, USA*
Advisors: Prof. Jiefeng Sun and Prof. Hyunglae Lee; CGPA: 4/4

Master of Technology in Mechanical Engineering — **Aug. 2021**
College of Engineering Pune, Savitribai Phule Pune University — *Pune, India*
Thesis: Synthesis and Fatigue Life Optimization of a Spring-less Compliant Robot Leg Design
Advisor: Prof. Ajay Bhattu; CGPA: 9.51/10

HONORS AND SELECTED AWARDS

Jumpstart Research Grant ($2,250 total) 2025–2026
Arizona State University (Spring and Fall 2025; Spring 2026)

GPSA Travel Grant ($950) Summer 2026
Graduate and Professional Student Association, Arizona State University

Junior Research Fellowship ($10,895.92) 2021–2023
Indian Institute of Technology Delhi

AICTE Post Graduate Scholarship ($3,386.20) 2019–2021
All India Council for Technical Education

Bravo Award, Garrett Motion 2021

Graduate Aptitude Test in Engineering (GATE), 98.22 percentile 2019

PUBLICATIONS

Journal Articles and Preprints

7. E. Weissman, **R. Khatavkar**, and J. Sun, *Versatile artificial muscles by decoupling anisotropy*, Proceedings of the National Academy of Sciences, vol. 123, no. 13, Art. no. e2529273123, 2026.

6. **R. Khatavkar**, T. B. Nguyen, I. Kang, H. Lee, and J. Sun, *Soft Semi-active Back Support Device with Adaptive Force Profiles using Variable-elastic Actuation and Weight Feedback*, arXiv preprint arXiv:2603.03724, 2026. [Preprint]

5. Y. Chen, **R. Khatavkar**, S. Nayak, J. Sun, and H. Lee, *Impact of a Soft Wearable Back-Support Device on Postural Stability during Trip-Like Perturbations*, arXiv preprint arXiv:2606.02888, 2026. [Preprint]

4. **R. Khatavkar**, J. Sun, and H. Lee, *Improved Postural Stability Using a Lightweight Semi-Active Soft Back Support Device Under Standing Perturbations*, arXiv preprint arXiv:2606.02928, 2026. [Preprint]

3. **R. Khatavkar**, T. B. Nguyen, Y. Chen, H. Lee, and J. Sun, *A Hybrid Variable-Stiffness Soft Back Support Device*, IEEE Robotics and Automation Letters, 2025. [DOI]

2. **R. Khatavkar**, A. Tiwari, P. Bhat, A. K. Srivastava, S. S. Kumaran, and D. Joshi, *A Novel Kinematic Gait Parameter-Based Vibrotactile Cue for Freezing of Gait Mitigation Among Parkinson's Patients: A Pilot Study*, IEEE Transactions on Haptics, vol. 17, no. 4, pp. 689–704, 2024. [DOI]

1. **R. Khatavkar**, A. Tiwari, P. Bhat, and D. Joshi, *Investigating the Effects of a Kinematic Gait Parameter-Based Haptic Cue on Toe Clearance in Parkinson's Patients*, Annals of Biomedical Engineering, vol. 52, no. 8, pp. 2039–2050, 2024. [DOI]

Conference Proceedings

4. **R. V. Khatavkar** and A. P. Bhattu, *Synthesis and Fatigue Life Optimization of a Spring-Less Compliant Robot Leg Design*, in National Conference on Multidisciplinary Analysis and Optimization, pp. 451–459, Springer, 2023. [DOI]
3. **R. Khatavkar**, A. Tiwari, R. Bajpai, and D. Joshi, *Gait Step Length Classification Using Force Myography*, in 2022 International Conference for Advancement in Technology (ICONAT), pp. 1–4, 2022. [DOI]
2. A. Tiwari, R. Bajpai, **R. Khatavkar**, M. Gupta, B. Garg, and D. Joshi, *Exploring the Center of Pressure Shift Feedback at Heel Strike to Modulate the Step Length*, in 2022 International Conference for Advancement in Technology (ICONAT), pp. 1–5, 2022. [DOI]
1. R. Bajpai, A. Tiwari, D. Joshi, and **R. Khatavkar**, *AbnormNet: A Neural Network Based Suggestive Tool for Identifying Gait Abnormalities in Cerebral Palsy Children*, in 2022 International Conference for Advancement in Technology (ICONAT), pp. 1–5, 2022. [DOI]

Patents

1. J. Sun, **R. Khatavkar**, and H. Lee, *Systems and Methods for a Hybrid Soft Variable Stiffness Back Support Device*, U.S. Provisional Patent Application Serial No. 63/912,143.

PROPOSAL DRAFTING EXPERIENCE

1. Contributed to the proposal led by Profs. Jiefeng Sun, Hyunglae Lee, and Daniel Peterson: *Artificial-Muscle-Driven Bidirectional Trunk Assistance for Improving Postural Balance.* **National Institutes of Health (NIH), R21 Trailblazer.** I provided 50% of the preliminary results and drafted 25% of the proposal.

INDUSTRIAL EXPERIENCES

R&D Intern, Garrett Motion **Aug.–Sep. 2021**

Torque-to-Yield Bolted Joints for High-Pressure Compressors

- Designed a torque-to-yield bolted joint to maintain joint integrity at high temperatures.

R&D Intern, American Axle and Manufacturing **Aug. 2017–May 2018**

Electronic Differential Locking Mechanism

- Integrated an electronic locking feature into an existing open differential while minimizing changes to the original assembly, and built a functional prototype.
- Developed an automatic locking algorithm in MATLAB/Simulink.

INVITED TALKS AND PRESENTATIONS

Invited Talks

1. *A Hybrid Variable-Stiffness Soft Back Support Device*, Arizona State University, Spring 2025.

Conference Presentations

2. *Impact of a Soft Wearable Back-Support Device on Postural Stability during Trip-Like Perturbations*, IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), oral presentation, 2026.

1. *Improved Postural Stability Using a Lightweight Semi-Active Soft Back Support Device Under Standing Perturbations*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), poster presentation, 2026.

MEDIA COVERAGE (SELECTED)

1. **ASU News:** Giving robots more muscle can help them lose weight (04/01/2026).

PROFESSIONAL ACTIVITIES

- **Journal Reviewer**
  - Scientific Reports; IEEE Sensors Journal; IEEE Robotics and Automation Letters; Journal of NeuroEngineering and Rehabilitation.
- **Conference Reviewer**
  - IEEE International Conference on Soft Robotics (RoboSoft); IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM); American Control Conference (ACC).
- **Professional Memberships**
  - Member, Tau Beta Pi; Student Member, IEEE.

SKILLS

- **Tools:** MATLAB, PyTorch, OpenSim, Arduino, Raspberry Pi, Vicon Nexus, SolidWorks, Fusion 360, Siemens NX, Abaqus CAE, Ansys, SMC PneuDraw, Overleaf, Adobe Illustrator, and Adobe Premiere Pro.
- **Design:** Mechatronic and robotic system design, pneumatic system design, CAD, additive manufacturing, solid mechanics, and finite-element analysis.
- **Experimental:** Instron force testing, Vicon optical motion capture, Delsys EMG acquisition and processing, Speedgoat real-time multi-sensor capture, movement biomechanics, signal processing, clinical protocol development, and participant testing.

MENTORING, LEADERSHIP AND ACTIVITIES

- **Student Research Mentor:** The Bach Nguyen (B.S.), Rayan Mostofa (B.S.), Cindy Furukawa (honors thesis), Akshay Karthik (high school; AzSEF 2024 award), and Alex Huang (high school; AzSEF 2024 award).
- **Graduate Admissions Assistance, College of Engineering Pune:** Assisted the Admissions Committee in evaluating applicants to the master's program in mechanical engineering.